72. What is the remainder when 51×27×35×62×75 is divided by 100?

(a) 50 (b) 25
(c) 5 (d) 1

Ans. (a) : $\frac{51\times27\times35\times62\times75}{100}$

Simplify by 25 (on dividing the numerator and denominator by 25)

$\Rightarrow \frac{51\times27\times35\times62\times3}{4}$

$\Rightarrow \frac{51}{4} \rightarrow$ Remainder = 3

$\Rightarrow \frac{27}{4} \rightarrow$ Remainder = 3

$\Rightarrow \frac{35}{4} \rightarrow$ Remainder = 3

$\Rightarrow \frac{62}{4} \rightarrow$ Remainder = 2

$\Rightarrow \frac{3}{4} \rightarrow$ Remainder = 3

then $\frac{3\times3\times3\times2\times3}{4}$

$\Rightarrow \frac{162}{4} \rightarrow$ Remainder = 2

$\therefore$ Required remainder - last quotient × Simplify

$= 2 \times 25$

$= 50$

73. A sum of ₹ 2,500 is distributed among X, Y and Z in the ratio $\frac{1}{2}:\frac{3}{4}:\frac{5}{6}$. What is the difference between the maximum share and the minimum share?

(a) ₹ 300 (b) ₹ 350
(c) ₹ 400 (d) ₹ 450

Ans. (c) : $\because$ $x:y:z=\frac{1}{2}:\frac{3}{4}:\frac{5}{6}$

$x:y:z=\frac{6}{12}:\frac{9}{12}:\frac{10}{12}$

$x:y:z=6:9:10$

The difference between the maximum share and minimum share-

$=2500\times\frac{(10-6)}{6+9+10}$

$=2500\times\frac{4}{25}$

$=100\times4=400$

74. For what value of n, the sum of digits in the number $(10^n + 1)$ is 2?

(a) For $n = 0$ only
(b) For any whole number n
(c) For any positive integer n only
(d) For any real number n

Ans. (b) : (10^n+1)

option (a) $n = 0$

$10^0+1 \Rightarrow 1+1 \Rightarrow 2$ (Sum of digits)

option (b) $\because$ n पूर्ण संख्या है,

$n = 0$

$10^0+1=1+1=2$ (Sum of digits)

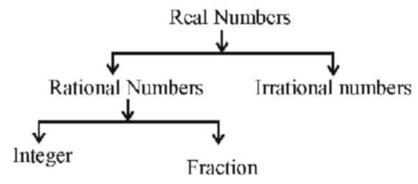
$10^1+1=10+1=2$ (Sum of digits)

$10^2+1 \Rightarrow 100+1 \Rightarrow 2$ (Sum of digits)

Option (c),

When the sum of the digits on whole numbers is 2 then It will be the same for positive integer numbers as well.

Option (d),



According to flow chart,

for real numbers,

The sum of $(10^n + 1)$ digits will not be 2.

Hence,

The option (c) "for any positive integer n only" is correct.

75. In a class, there are three groups A, B and C. If one student from group A and two students from group B are shifted to group C, then what happens to the average weight of the students of the class?

(a) It increases
(b) It decreases
(c) It remains the same.
(d) No conclusion can be drawn due to insufficient data.

Ans. (c) : The average weight of the students of the class will remain the same as the number of students in the class is the same, only the students have been transferred in groups.

76. How many different sums can be formed with the denominations ₹ 50, ₹ 100, ₹ 200, ₹ 500 and ₹ 2,000 taking at least three denominations at a time?

(a) 16 (b) 15
(c) 14 (d) 10

Ans. (a) : Required number $= 5_{C_3} + 5_{C_4} + 5_{C_5}$

$=\frac{5!}{3!\times(5-3)!}+\frac{5!}{4!\times(5-4)!}+\frac{5!}{5!(5-5)!}$

$\left\{\because\ {}_{n}C_{r}=\frac{n!}{r!(n-r)!}\right\}$

$=\frac{5!}{3!\times2!}+\frac{5!}{4!\times1!}+\frac{5!}{5!\times0!}$

$=\frac{5\times4\times3!}{3!\times2\times1}+\frac{5\times4!}{4!\times1}+\frac{5!}{5!}$ $[\because\ 0!=1]$

$= 10 + 5 + 1 = 16$

77. A shop owner offers the following discount option on an article to a customer:

1. **Successive discounts of 10% and 20%, and then pay a service tax of 10%**
2. **Successive discounts of 20% and 10%, and then pay a service tax of 10%**
3. **Pay a service tax of 10% first, then successive discounts of 20% and 10%**

Which one of the following is correct?

(a) 1 only is the best option for the customer.
(b) 2 only is the best option for the customer.
(c) 3 only is the best option for the customer
(d) All the options are equally good for the customer.

Ans. (d) : Let lebelled price = ₹ 100

from statement (I)

First discount = 10% = ₹ 90

Second discount = 20% $=90\times\frac{80}{100}\Rightarrow$ ₹ 72

Service Tax = 10% $=72\times\frac{90}{100}\Rightarrow$ ₹ 64.8

From statement-(ii)

First discount = 20% $\Rightarrow$ ₹ 80

Second discount = 10% $\Rightarrow 80\times\frac{90}{100}=$ ₹ 72

Service Tax = 10% $=72\times\frac{90}{100}=$ ₹ 64.8

From statement-(iii)

Service Tax = 10% = ₹ 90

First discount = 20% $\Rightarrow 90\times\frac{80}{100}=$ ₹ 72

Second discount = 10% $=72\times\frac{90}{100}=$ ₹ 64.8

Hence, All the options are equally good for the customers.

78. The letters from A to Z are numbered from 1 to 26 respectively. If GHI = 1578 and DEF = 912, then what is ABC equal to

(a) 492 (b) 468
(c) 262 (d) 246

Ans. (d) : $\because$ G H I = 7 8 9 (From Alphabetically order)

then G H I = 789 × 2

= 1578

D E F = 4 5 6 (From Alphabetically order)

= 456 × 2

= 912

Similarly

A B C = 1 2 3 (From Alphabetically order)

= 123 × 2

ABC = 246

79. What is the missing term @ in the following?

ACPQ : BESU :: MNGI : @

(a) NPJL (b) NOJM
(c) NPIL (d) NPJM

Ans. (d) : Just as,

1	3	16	17
A	C	P	Q
+1↓	+2↓	+3↓	+4↓
B	E	S	U
2	5	19	21

Similarly

13	14	7	9
M	N	G	I
+1↓	+2↓	+3↓	+4↓
N	P	J	M
14	16	10	13

80. What is the largest number among the following?

(a) $\left(\frac{1}{2}\right)^{-6}$ (b) $\left(\frac{1}{4}\right)^{-3}$
(c) $\left(\frac{1}{3}\right)^{-4}$ (d) $\left(\frac{1}{6}\right)^{-2}$

Ans. (c) : $\left(\frac{1}{2}\right)^{-6} \Rightarrow \frac{1}{2^{-6}} \Rightarrow 2^6 \Rightarrow 64$

$\left(\frac{1}{4}\right)^{-3} \Rightarrow \frac{1}{4^{-3}} \Rightarrow 4^3 = 64$

$\left(\frac{1}{3}\right)^{-4} \Rightarrow \frac{1}{3^{-4}} \Rightarrow 3^4 \Rightarrow 81$

$\left(\frac{1}{6}\right)^{-2} \Rightarrow \frac{1}{6^{-2}} = 6^2 = 36$

Hence, The largest number $=\left(\frac{1}{3}\right)^{-4}$